Analytical Chemistry Cumulative Exam
Friday, November 21, 2008

I. Warm-up (30 points).

1) The pH of a 1.6 M HNO_3 solution is: _____.

2) The solubility of $Mg(OH)_2$ (K_{sp}= 7.1 x 10^{-12}) will (increase/decrease)_____ in a solution with a pH of 12 as compared to a solution with a pH of 2.

3) (True/False) _____ The solubility of saturated AgI (K_{sp}= 8.3 x 10^{-17}) will decrease in a 0.1 M $NaNO_3$ solution as compared to de-ionized water.

4) The solubility of AgCl (Ksp = 1.8 x 10^{-10}) in a 0.03 M $NaNO_3$ solution will be (greater than, less than, equal to) _____ the solubility of AgCl in a 0.01 M $Mg(ClO_4)_2$ solution.

5) _____(True/ False) The pH of 0.1M HCl in a 1.0M NaCl solution equals the pH of 0.1 M HCl in a 0.01 M NaCl solution. (Assume a standard glass electrode was used for this measurement.)

6) Write the charge balance for a solution containing 0.1 M phthalic acid, a diacid, (abbreviate H_2Phth in your answer) in distilled, deionized water.

II. Problems.

7. (35 points) Consider the following aqueous equilibrium for the copper salt shown below:

$Cu_4(OH)_6(SO_4)$ (s) $\leftrightarrows$ $4Cu^{2+}$ (aq) + $6OH^-$ (aq) + SO_4^{2-} (aq) $\qquad K_{sp} = 2.3 \times 10^{-69}$

Find the concentration (M) of Cu^{2+}(aq) if the solution pH is fixed to a value of 2.

8. (35 points) Find the pH of a 50 mM (formal concentration) sodium phosphate buffer (pH 7.0) with an initial volume of 10 mL to which 1.0 mL of 1.0 M HCl has been added. The pKa values for phosphoric acid are: 2.15, 7.2, and 12.4.